

Disable or Modify Tools

Sub-techniques (6)

Adversaries may disable, degrade, or tamper with security tools or applications (e.g., endpoint detection and response (EDR) tools, intrusion detection systems (IDS), antivirus, logging agents, sensors, etc.) to impair or reduce visibility of defensive capabilities. This may include stopping specific services, killing processes, modifying or deleting tool configuration files and Registry keys, or preventing tools from updating. This may also include impairing defenses more broadly by disrupting preventative, detection, and response mechanisms across host, network, and cloud environments.^[1]

In addition to directly targeting tools, adversaries may block or manipulate indicators and telemetry used for detection. This includes maliciously disabling or redirecting sensors such as Event Tracing for Windows (ETW), modifying event log configurations (e.g., redirecting Security logs), or interfering with logging pipelines and forwarding mechanisms (e.g., SIEM ingestion).^{[2][3]}

More advanced techniques include leveraging legitimate drivers or debugging mechanisms to render tools non-functional, bypassing anti-tampering protections, and targeting specific defenses such as Sysmon or cloud monitoring agents. Adversaries may also disrupt broader defensive operations, including update mechanisms, logging infrastructure (e.g., syslog), or event aggregation, further degrading an organization’s ability to detect and respond to malicious activity.^[4]

ID: T1685

Sub-techniques: [T1685.001](#), [T1685.002](#), [T1685.003](#), [T1685.004](#), [T1685.005](#), [T1685.006](#)

-  **Tactic:** [Defense Impairment](#)
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Procedure Examples

ID	Name	Description
C0028	2015 Ukraine Electric Power Attack	During the 2015 Ukraine Electric Power Attack , Sandworm Team modified in-registry internet settings to lower internet security. ^[5]
S0331	Agent Tesla	Agent Tesla has the capability to kill any running analysis processes and AV software. ^[6]
G1030	Agrius	Agrius used several mechanisms to try to disable security tools. Agrius attempted to modify EDR-related services to disable auto-start on system reboot. Agrius used a publicly available driver, <code>GMEr64.sys</code> typically used for anti-rootkit functionality, to selectively stop and remove security software processes. ^[7]
G1024	Akira	Akira has disabled or modified security tools for defense evasion. ^[8]
G0082	APT38	APT38 has unhooked DLLs to disable endpoint detection and response (EDR) or anti-virus (AV) tools. ^[9]
G0096	APT41	APT41 developed a custom injector that enables an Event Tracing for Windows (ETW) bypass, making malicious processes invisible to Windows logging. ^[10]
G1023	APT5	APT5 has used the CLEANPULSE utility to insert command line strings into a targeted process to prevent certain log events from occurring. ^[11]
G0143	Aquatic Panda	Aquatic Panda has attempted to stop endpoint detection and response (EDR) tools on compromised systems. ^[12]
C0046	ArcaneDoor	ArcaneDoor modified the Authentication, Authorization, and Accounting (AAA) function of targeted Cisco ASA appliances to allow the threat actor to bypass normal AAA operations. ^{[13][14]}
S0640	Avaddon	Avaddon looks for and attempts to stop anti-malware solutions. ^[15]
S0638	Babuk	Babuk can stop anti-virus services on a compromised host. ^[16]
S0534	Bazar	Bazar has manually loaded ntdll from disk in order to identify and remove API hooks set by security products. ^[17]
G1043	BlackByte	BlackByte disabled security tools such as Windows Defender and the Raccine anti-ransomware tool during operations. ^{[18][19][20]}
S1180	BlackByte Ransomware	BlackByte Ransomware adds .JS and .EXE extensions to the Microsoft Defender exclusion list. BlackByte Ransomware terminates and removes the Raccine anti-ransomware utility. ^[21]
S1184	BOLDMOVE	BOLDMOVE can disable the Fortinet daemons <code>moglogd</code> and <code>syslogd</code> to evade detection and logging. ^[22]
S0252	Brave Prince	Brave Prince terminates antimalware processes. ^[23]
G0060	BRONZE BUTLER	BRONZE BUTLER has incorporated code into several tools that attempts to terminate anti-virus processes. ^[24]
S1063	Brute Ratel C4	Brute Ratel C4 has the ability to hide memory artifacts and to patch Event Tracing for Windows (ETW) and the Anti Malware Scan Interface (AMSI). ^{[25][26]}
S0482	Bundlore	Bundlore can change browser security settings to enable extensions to be installed. Bundlore uses the <code>pkill cfprefsd</code> command to prevent users from inspecting processes. ^{[27][28]}
S0484	Carberp	Carberp has attempted to disable security software by creating a suspended process for the security software and injecting code to delete antivirus core files when the process is resumed. ^[29]

ID	Name	Description
S0608	Conficker	Conficker terminates various services related to system security and Windows. ^[34]
G1052	Contagious Interview	Contagious Interview has convinced victims to disable Docker and other container environments and run code on their machine natively in attempts to bypass container isolation and ensure device infection. ^[35]
C0029	Cutting Edge	During Cutting Edge , threat actors disabled logging and modified the <code>compcheckresult.cgi</code> component to edit the Ivanti Connect Secure built-in Integrity Checker exclusion list to evade detection. ^{[36][37]}
S0334	DarkComet	DarkComet can disable Security Center functions like anti-virus. ^{[38][39]}
S1111	DarkGate	DarkGate will terminate processes associated with several security software products if identified during execution. ^[40]
S9017	DCRAT	DCRAT can patch Microsoft's Antimalware Scan Interface (AMSI) to evade detection. ^[41]
S0659	Diavol	Diavol can attempt to stop security software. ^[42]
S0695	Donut	Donut can patch Antimalware Scan Interface (AMSI), Windows Lockdown Policy (WLDP), as well as exit-related Native API functions to avoid process termination. ^[43]
S9013	DRYHOOK	DRYHOOK has killed all instances of the <code>cgi-server</code> process in order for the modified Perl module to be activated. ^[44]
S0377	Ebury	Ebury can disable SELinux Role-Based Access Control and deactivate PAM modules. ^[45]
S0554	Egregor	Egregor has disabled Windows Defender to evade protections. ^[46]
S0605	EKANS	EKANS stops processes related to security and management software. ^{[47][48]}
G0037	FIN6	FIN6 has deployed a utility script named <code>kill.bat</code> to disable anti-virus. ^[49]
G0047	Gamaredon Group	Gamaredon Group has delivered macros which can tamper with Microsoft Office security settings. ^{[50][51]}
S0249	Gold Dragon	Gold Dragon terminates anti-malware processes if they're found running on the system. ^[23]
S0477	Goopy	Goopy has the ability to disable Microsoft Outlook's security policies to disable macro warnings. ^[52]
G0078	Gorgon Group	Gorgon Group malware can attempt to disable security features in Microsoft Office and Windows Defender using the <code>taskkill</code> command. ^[53]
S0531	Grandoreiro	Grandoreiro can hook APIs, kill processes, break file system paths, and change ACLs to prevent security tools from running. ^[54]
S0132	H1N1	H1N1 kills and disables services for Windows Security Center, and Windows Defender. ^[55]
S0061	HDoor	HDoor kills anti-virus found on the victim. ^[56]
S0697	HermeticWiper	HermeticWiper has the ability to set the <code>HKLM:\SYSTEM\CurrentControlSet\Control\CrashControl\CrashDumpEnabled</code> Registry key to <code>0</code> in order to disable crash dumps. ^{[57][58][59]}
S0601	Hildegard	Hildegard has modified DNS resolvers to evade DNS monitoring tools. ^[60]
C0038	HomeLand Justice	During HomeLand Justice , threat actors modified and disabled components of endpoint detection and (EDS) to prevent detection. Microsoft Security Center (MSC) was also disabled. ^[61]

ID	Name	Description
G1032	INC Ransom	INC Ransom can use SystemSettingsAdminFlows.exe, a native Windows utility, to disable Windows Defender. ^[64]
G0119	Indrik Spider	Indrik Spider used PsExec to leverage Windows Defender to disable scanning of all downloaded files and to restrict real-time monitoring. ^[65] Indrik Spider has used <code>MpCmdRun</code> to revert the definitions in Microsoft Defender. ^[66] Additionally, Indrik Spider has used WMI to stop or uninstall and reset anti-virus products and other defensive services. ^[66]
S0201	JPIN	JPIN can lower security settings by changing Registry keys. ^[67]
S1206	JumbledPath	JumbledPath can impair logging on all devices used along its connection path to compromised hosts. ^[68]
G0094	Kimsuky	Kimsuky has been observed turning off Windows Security Center and can hide the AV software window from the view of the infected user. ^{[69][70]}
S0669	KOCTOPUS	KOCTOPUS will attempt to delete or disable all Registry keys and scheduled tasks related to Microsoft Security Defender and Security Essentials. ^[71]
C0035	KV Botnet Activity	KV Botnet Activity used various scripts to remove or disable security tools, such as <code>http_watchdog</code> and <code>firewallsd</code> , as well as tools related to other botnet infections, such as <code>mips_ff</code> , on victim devices. ^[72]
G0032	Lazarus Group	Lazarus Group malware TangoDelta attempts to terminate various processes associated with McAfee. Additionally, Lazarus Group malware SHARPKNOT disables the Microsoft Windows System Event Notification and Alerter services. ^{[73][74][75][76]}
S9039	LazyWiper	LazyWiper can disable Microsoft Windows Defender Real-Time Monitoring with the <code>Set-MpPreference</code> cmdlet. ^[77]
S1199	LockBit 2.0	LockBit 2.0 can disable firewall rules and anti-malware and monitoring software including Windows Defender. ^{[78][79]}
S1202	LockBit 3.0	LockBit 3.0 can disable security tools to evade detection including Windows Defender. ^{[80][81][82]}
S0372	LockerGoga	LockerGoga installation has been immediately preceded by a "task kill" command in order to disable anti-virus. ^[83]
S1213	Lumma Stealer	Lumma Stealer has attempted to bypass Windows Antimalware Scan Interface (AMSI) by removing the string "AmsiScanBuffer" from the "clr.dll" module in memory to prevent it from being called. ^[84]
S1048	macOS.OSAMiner	macOS.OSAMiner has searched for the Activity Monitor process in the System Events process list and kills the process if running. macOS.OSAMiner also searches the operating system's <code>install.log</code> for apps matching its hardcoded list, killing all matching process names. ^[85]
G0059	Magic Hound	Magic Hound has disabled antivirus services on targeted systems in order to upload malicious payloads. ^[86]
S1169	Mango	Mango contains an unused capability to block endpoint security solutions from loading user-mode code hooks via a DLL in a specified process by using the <code>UpdateProcThreadAttribute</code> API to set the <code>PROC_THREAD_ATTRIBUTE_MITIGATION_POLICY</code> to <code>PROCESS_CREATION_MITIGATION_POLICY_BLOCK_NON_MICROSOFT_BINARIES_ALWAYS_ON</code> for an identified process. ^[87]
S0449	Maze	Maze has disabled dynamic analysis and other security tools including IDA debugger, x32dbg, and OllyDbg. ^[88] It has also disabled Windows Defender's Real-Time Monitoring feature and attempted to disable endpoint protection services. ^[89]

ID	Name	Description
S0576	MegaCortex	MegaCortex was used to kill endpoint security processes. ^[94]
S0455	Metamorfo	Metamorfo has a function to kill processes associated with defenses and can prevent certain processes from launching. ^{[95][96]}
S0688	Meteor	Meteor can attempt to uninstall Kaspersky Antivirus or remove the Kaspersky license; it can also add all files and folders related to the attack to the Windows Defender exclusion list. ^[97]
G1054	MirrorFace	MirrorFace has disabled Windows Defender in compromised environments. ^[98]
G0069	MuddyWater	MuddyWater can disable the system's local proxy settings. ^[99]
S1135	MultiLayer Wiper	MultiLayer Wiper removes the Volume Shadow Copy (VSS) service from infected devices along with all present shadow copies. ^[7]
S0228	NanHaiShu	NanHaiShu can change Internet Explorer settings to reduce warnings about malware activity. ^[100]
S0336	NanoCore	NanoCore can modify the victim's anti-virus. ^{[101][102]}
S0457	Netwalker	Netwalker can detect and terminate active security software-related processes on infected systems. ^{[103][104]}
C0002	Night Dragon	During Night Dragon , threat actors disabled anti-virus and anti-spyware tools in some instances on the victim's machines. The actors also disabled proxy settings to allow direct communication from victims to the Internet. ^[105]
S9014	PHASEJAM	PHASEJAM has modified Ivanti Connect Secure appliances and blocks the system upgrades by altering the DSUpgrade.pm file. ^[44]
G1040	Play	Play has used tools including GMER, IOBit, and PowerTool to disable antivirus software. ^{[106][107]}
S0223	POWERSTATS	POWERSTATS can disable Microsoft Office Protected View by changing Registry keys. ^[108]
S0279	Proton	Proton kills security tools like Wireshark that are running. ^[109]
S9019	PureCrypter	PureCrypter has executed <code>Set-MpPreference -ExclusionPath</code> to exclude files or folders from Windows Defender scans. ^[110]
G0024	Putter Panda	Malware used by Putter Panda attempts to terminate processes corresponding to two components of Sophos Anti-Virus (SAVAdminService.exe and SavService.exe). ^[111]
S0583	Pysa	Pysa has the capability to stop antivirus services and disable Windows Defender. ^[112]
S0650	QakBot	QakBot has the ability to modify the Registry to add its binaries to the Windows Defender exclusion list. ^[113]
S1242	Qilin	Qilin can terminate antivirus-related processes and services. ^{[114][115][116][117]}
C0055	Quad7 Activity	Quad7 Activity has disabled the TP-Link management interface for TP-Link by killing the <code>/usr/bin/httpd</code> process. ^{[118][119][120]}
S0481	Ragnar Locker	Ragnar Locker has attempted to terminate/stop processes and services associated with endpoint security products. ^[121]
S1130	Raspberry Robin	Raspberry Robin can add an exception to Microsoft Defender that excludes the entire main drive from anti-malware scanning to evade detection. ^[122]

ID	Name	Description
G0106	Rocke	Rocke used scripts which detected and uninstalled antivirus software. ^{[126][127]}
S0253	RunningRAT	RunningRAT kills antimalware running process. ^[23]
S0446	Ryuk	Ryuk has stopped services related to anti-virus. ^[128]
G1031	Saint Bear	Saint Bear will modify registry entries and scheduled task objects associated with Windows Defender to disable its functionality. ^[129]
G1015	Scattered Spider	Scattered Spider has uninstalled and disabled security tools. ^[130]
S9008	Shai-Hulud	Shai-Hulud has replaced DNS configuration from <code>/tmp/resolved.conf</code> in order to gain control of network-level control within CI environments and has flushed iptables rules using <code>sudo iptables -F OUTPUT</code> and <code>sudo iptables -F DOCKER-USER</code> . ^[131]
C0058	SharePoint ToolShell Exploitation	During SharePoint ToolShell Exploitation , threat actors disabled Microsoft Defender through Registry settings and real-time monitoring via PowerShell. ^{[132][133]}
S1178	ShrinkLocker	ShrinkLocker disables protectors used to secure the BitLocker encryption key on victim systems. ^{[134][135]}
S0692	SILENTRINITY	SILENTRINITY 's <code>amsiPatch.py</code> module can disable Antimalware Scan Interface (AMSI) functions. ^[136]
S0468	Skidmap	Skidmap has the ability to set SELinux to permissive mode. ^[137]
C0024	SolarWinds Compromise	During the SolarWinds Compromise , APT29 used the service control manager on a remote system to disable services associated with security monitoring products. ^[138]
S9024	SPAWNCHIMERA	SPAWNCHIMERA has modified the Ivanti Integrity Checker Tool to evade detection. ^{[139][140]}
S1234	SplatCloak	SplatCloak has identified and disabled API callback features of Windows Defender and Kaspersky. ^[141]
S0058	SslIMM	SslIMM identifies and kills anti-malware processes. ^[56]
S1200	StealBit	StealBit can configure processes to not display certain Windows error messages by through use of the <code>NtSetInformationProcess</code> . ^[142]
S0491	StrongPity	StrongPity can add directories used by the malware to the Windows Defender exclusions list to prevent detection. ^[143]
S0603	Stuxnet	Stuxnet reduces the integrity level of objects to allow write actions. ^[144]
S0559	SUNBURST	SUNBURST attempted to disable software security services following checks against a FNV-1a + XOR hashed hardcoded blacklist. ^[145]
G1018	TA2541	TA2541 has attempted to disable built-in security protections such as Windows AMSI. ^[146]
G0092	TA505	TA505 has used malware to disable Windows Defender. ^[147]
G0139	TeamTNT	TeamTNT has disabled and uninstalled security tools such as Alibaba, Tencent, and BMC cloud monitoring agents on cloud-based infrastructure. ^{[148][149]}
S0595	ThiefQuest	ThiefQuest uses the function <code>kill_unwanted</code> to obtain a list of running processes and kills each process matching a list of security related processes. ^[150]
S0004	ThiefQuest	ThiefQuest kills the Anti-Malware service. ^[151]

ID	Name	Description
G1048	UNC3886	UNC3886 has disabled OpenSSL digital signature verification of system files through corruption of boot files. ^[154]
S0130	Unknown Logger	Unknown Logger has functionality to disable security tools, including Kaspersky, BitDefender, and MalwareBytes. ^[155]
G1047	Velvet Ant	Velvet Ant attempted to disable local security tools and endpoint detection and response (EDR) software during operations. ^[156]
S0670	WarzoneRAT	WarzoneRAT can disarm Windows Defender during the UAC process to evade detection. ^[157]
S0579	Waterbear	Waterbear can hook the <code>ZwOpenProcess</code> and <code>GetExtendedTcpTable</code> APIs called by the process of a security product to hide PIDs and TCP records from detection. ^[158]
S0689	WhisperGate	WhisperGate can download and execute AdvancedRun.exe to disable the Windows Defender Threat Protection service and set an exclusion path for the C:\ drive. ^{[159][160][161]}
G0102	Wizard Spider	Wizard Spider has shut down or uninstalled security applications on victim systems that might prevent ransomware from executing. ^{[162][163][164][165]}
S1065	Woody RAT	Woody RAT has suppressed all error reporting by calling <code>SetErrorMode</code> with 0x8007 as a parameter. ^[166]
S1207	XLoader	XLoader loads a copy of NTDLL to evade hooks from security monitoring tools on this library. ^[167] XLoader can add the path of its executable to the Microsoft Defender exclusion list. ^[168]
S1114	ZIPLINE	ZIPLINE can add itself to the exclusion list for the Ivanti Connect Secure Integrity Checker Tool if the <code>--exclude</code> parameter is passed by the <code>tar</code> process. ^[169]
S0412	ZxShell	ZxShell can kill AV products' processes. ^[170]

Mitigations

ID	Mitigation	Description
M1047	Audit	Periodically verify that tools are functioning appropriately – for example, that all expected hosts with EDRs or monitoring agents are checking in to the central console. Check EDRs to ensure that no unexpected exclusion paths have been added. In Microsoft Defender for Endpoint, exclusions can be reviewed with the <code>Get-MpPreference</code> cmdlet. ^[171]
M1042	Disable or Remove Feature or Program	Consider removing previous versions of tools that are unnecessary to the environment when possible.
M1038	Execution Prevention	Use application control where appropriate, especially regarding the execution of tools outside of the organization's security policies (such as rootkit removal tools) that have been abused to impair system defenses. Ensure that only approved security applications are used and running on enterprise systems.
M1022	Restrict File and Directory Permissions	Ensure proper process and file permissions are in place to prevent adversaries from disabling or interfering with security services.
M1024	Restrict Registry Permissions	Ensure proper Registry permissions are in place to prevent adversaries from disabling or interfering with security services.
M1054	Software	Consider automatically relaunching forwarding mechanisms at recurring intervals (ex: temporal, on login, etc.)

Detection Strategy

ID	Name	Analytic ID	Analytic Description
DET0497	Detection of Defense Impairment through Disabled or Modified Tools across OS Platforms.	AN1369	Detection of adversary behavior that disables or modifies security tools, including killing AV/EDR processes, stopping services, altering Sysmon registry keys, or tampering with exclusion lists. Defenders observe process/service termination, registry modification, and abnormal absence of expected telemetry.
		AN1370	Detects kill/systemctl/service commands against EDR, auditd, falco, osquery, rsyslog, journald, or agent processes; configuration edits disabling startup; module unload attempts; abrupt cessation of logs after privileged shell execution.
		AN1371	Detection of adversary disabling endpoint security tools by unloading launch agents/daemons, modifying configuration profiles, or disabling Gatekeeper/XProtect/logging settings, or removing endpoint agents followed by telemetry loss.
		AN1372	Correlates control-plane API actions disabling cloud-native monitoring or sensor agents (CloudTrail, GuardDuty, Security Hub, Defender, monitoring agents), role abuse preceding disablement, or instance agent uninstall events
		AN1373	Detects disabling container runtime security controls, removing sidecar sensors, modifying seccomp/AppArmor profiles, mounting host proc/sys paths to interfere with host logging, or killing in-container monitoring agents.
		AN1374	Detects disabling AAA, syslog, SNMP traps, ACL logging, or security features on routers/switches/firewalls; correlates privileged login followed by configuration commit reducing visibility.
		AN2044	Detects esxcli commands disabling syslog, firewall, lockdown mode, or stopping hostd/vpxa; correlates command execution with reduced forwarding activity.

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